

R3 Top Backup Roll Balance - System Primer

A bounded component-and-energy primer based on the words used in the preliminary incident record. It avoids asserting unverified design details.

Document ID	Incident	Record type	Status
R3-DEF-002	R3 carrier beam - 10/14/2024	Application test evidence	Synthetic / controlled

Known relationship map

Entity	Relationship stated or implied by the preliminary record	Why it matters	Status
Top backup roll balance	Was left isolated in the raised position after roll removal.	May define mechanical position and gravitational potential during outage work.	Historical source fact
B hydraulic system	Was shut off and bled for work on that system.	Pressure removal may alter support or restraint; the actual circuit boundary is separate evidence.	Historical source fact
Carrier beam and link	Fell together; both were reported intact after the event.	Fracture is not apparent, but seating, motion, wear, and retention still require examination.	Historical source fact
Keeper	Was reported still installed on the recovered assembly.	Presence does not by itself establish its design function, engagement, or load path.	Historical source fact
Yoke	Was the location on which the link rested; no apparent damage was reported.	Geometry, contact marks, clearances, and permitted movement remain relevant.	Historical source fact
Window liner	A small liner fell from the top operator side with the carrier beam.	Its timing, contact, and possible role are not established.	Historical source fact

Energy and configuration domains

Domain	Questions this primer should enable
Gravity	What mass was elevated, what normally supported it, and what positive restraint was required?
Hydraulic	Which circuit held or moved the balance, what was isolated, and how was zero pressure verified?
Mechanical	How were the beam, link, keeper, yoke, liner, and rolls positioned and retained?
Outage configuration	What changed when work and backup rolls were removed, and which temporary supports were required?

Required source hierarchy

- Use controlled drawings and equipment-specific procedures for design intent.
- Use signed field records, photos, video, and measurements for as-found state.
- Treat witness recollection as attributed evidence with visibility limits.
- Keep design requirements separate from inferred causal explanations.